

Enrichment Agent — Testing & Verification Guide

Date: 2026-08-18 · **Covers:** `docs/external-enrichment-agent.md` Phases 1–5

Every example below was run against the live dev data and the real public APIs — the expected outputs are what actually came back, not what should theoretically happen. Two ways to test: the terminal (fast, no calendar needed) and a real Outlook meeting (true end-to-end).

0. Setup, once

```
# 1. Create the cache table – DDL is applied by hand (see CLAUDE.md).
#   Paste supabase/enrichment-setup.sql into the DEV project's SQL editor.
#   Until this is run the agent still works; it just caches in-process only
#   and logs one warning.

# 2. Confirm the environment
#   SUPABASE_ENV=development → the 21,274-physician dev directory
#   ANTHROPIC_API_KEY set    → the paid web tier is available

npm run dev          # http://localhost:3002
```

Sign in at `/auth/login` before testing anything through the UI.

1. Fast path — test every case from the terminal

```
npm run enrich:try -- <email> [--free|--web] [--state NC] [--refresh] [--html]
```

`--free` = free registries only (no cost). `--web` forces the paid lookup. `--refresh` ignores the cache. `--html` prints the rendered brief instead of the summary.

The seven outcomes

#	Command	Expected STATUS	What it proves
1	<code>npm run enrich:try - - afrost@penmedicine.u penn.edu</code>	<code>in_bis</code> · 100 · 0 ms	Already in BIS → no enrichment, standard brief
2	<code>npm run enrich:try - - lisa.gangarosa@unchea lth.org --free</code>	<code>recovered_in_bis</code> · 90	An email miss is not a physician miss
3	<code>npm run enrich:try - - nshaheen@med.unc.edu --state NC --free</code>	<code>external</code> · 71	Free tiers alone identify someone outside BIS
4	<code>npm run enrich:try - - nshaheen@med.unc.edu --web --refresh</code>	<code>external</code> · 98	The paid tier, with proof URLs (~40 s)
5	<code>npm run enrich:try - - info@unch.unc.edu</code>	<code>facility_only</code> · 90 · < 0.5 s	Shared mailbox → facility, and no money spent
6	<code>npm run enrich:try - - abdul@primathon.in</code>	<code>not_physician</code>	Stops before any registry call
7	<code>npm run enrich:try - - zzqxx@nowhere- clinic-xyz.com -- free</code>	<code>unresolved</code>	Refuses to invent a person

Case 2 in full — the one that matters most

```
$ npm run enrich:try -- lisa.gangarosa@unchealth.org --free

STATUS      recovered_in_bis   (confidence 90)
NPI          1508935800
TIERS        T0:miss → T2:nppes-search → T3:recovered-by-npi → T4:pubmed → T4:cms-affiliation
BIS MATCH    Lisa M Gangarosa - Gastroenterology
BIS FACILITY UNC Hospitals Chapel Hill North Carolina [HSOP105211]
COLLEAGUES   Julia W Tang, Andrew J Gilman, Animesh Jain, Craig Reed, Neil D Shah
```

She **is** in `bis_physicians` as `lisa.gangarosa@unch.unc.edu` . Invited from a different address, the old code saw nothing. Now the agent resolves her NPI from the registry and recovers the whole BIS record. No AI, no cost.

Case 3 vs 4 — why the paid tier exists

```
npm run enrich:try -- nshaheen@med.unc.edu --free # unresolved
npm run enrich:try -- nshaheen@med.unc.edu --free --state NC # external, 71%
npm run enrich:try -- nshaheen@med.unc.edu --web --refresh # external, 98% + proof URLs
```

Without a state hint there are twenty Shaheens across the US and the agent correctly refuses to pick one. Give it the state and it lands on **NPI 1467521757**. The web tier supplies that hint on its own, by reading his UNC faculty page — and cites the pages that prove it, including one printing the email address verbatim.

Verify the provenance rendering

```
npm run enrich:try -- nshaheen@med.unc.edu --html > /tmp/brief.html && open /tmp/brief.html
```

Check for: the ⚠️ *not in your BIS database* banner, the green **Facility found in BIS** banner, a badge (🟢/🟡/🔴) and 🔗 on every row, the + **EXTRA — not held in BIS** block, and **Source disagreements** listing where BIS/NPPES overrode the web.

2. Real path — create meetings in Outlook

Book these in the calendar the app is signed in to. The rep's own address is the **organizer** in every case; that is deliberate — it is what proves the organizer rule.

Test A — physician already in BIS (baseline)

Subject	Lumendi DiLumen — Dr Frost
Attendee	afrost@penndmedicine.upenn.edu

Expect: the meeting expands to the normal physician block (brief, email intelligence, notes, actions). **No** enrichment card — nothing was enriched.

Test B — email missing from BIS, physician is not

Subject	UNC GI catch-up
Attendee	lisa.gangarosa@unchealth.org

Expect: an *External lookup* card resolving to **Lisa M Gangarosa**, `recovered_in_bis`, linked to **HSOP105211** with her five colleagues. Within five minutes the ingest tick also emails the normal brief and links the meeting to her NPI — check `[ingest] instant brief sent` in the server log.

Test C — genuinely outside BIS

Subject	DiLumen intro – UNC Hospitals GI
Attendee	nshaheen@med.unc.edu

Expect: the card appears within a few seconds from the free tiers. Because the address alone is ambiguous, press 🔍 **Identify with web search** — after ~40 s it resolves to Nicholas Shaheen at 98% with clickable proof URLs, the facility matched to HSOP105211, and Extra Intelligence showing industry payments, publications and trials.

The ingest tick separately emails a 🔍 **Outside BIS: ...** brief. Check the server log for `[ingest] external brief sent`.

Test D — the organizer must never be matched

Subject	Internal sync
Attendees	your own address only (you are also the organizer)

Expect: *no* enrichment card, *no* brief, no paid call. Then add a physician:

Subject	UNC GI + internal
Attendees	your own address and nshaheen@med.unc.edu

Expect: exactly **one** card — for the physician. Your own address is skipped even though it is on the attendee list. Add a meeting room to the invite and confirm it is skipped too.

Test E — title and description as context

Subject	Lumendi demo – Cleveland Clinic Mayfield Hts
Description	ESD volumes discussion
Attendee	anyone not in BIS

Expect: the facility resolves from the *title* to **HSOP101167 – Cleveland Clinic Mayfield Heights Ohio** and is passed to the lookup as a hint. Confirm the reverse too: put a **person's** name in the title and check it is *not* used to identify anyone — identity only ever comes from the attendee's email address.

Test F — shared mailbox

Subject	UNC Hospitals – scheduling
Attendee	info@unch.unc.edu

Expect: `facility_only` almost instantly (the domain index is built on first use, then it is sub-millisecond), facility HSOP105211 plus its physicians, and **no** paid call — the agent recognises an organisational address.

3. Verify the rules explicitly

Cost guards — confirm money is only spent when it must be

```
npm run enrich:try -- nshaheen@med.unc.edu --free      # PAID LOOKUP line absent
npm run enrich:try -- info@unch.unc.edu                # absent – shared mailbox
npm run enrich:try -- --name "Nicholas Shaheen" --state NC # absent – name supplied
npm run enrich:try -- nshaheen@med.unc.edu --web --refresh # present: N searches, tokens
```

Only the last one prints a `PAID LOOKUP` line. In the UI, simply opening a meeting never spends — the web tier is behind the button.

Cache

```
npm run enrich:try -- nshaheen@med.unc.edu --web --refresh # ~40 s
npm run enrich:try -- nshaheen@med.unc.edu                # ~0 ms, "from cache"
npm run enrich:try -- nshaheen@med.unc.edu --refresh      # full run again
```

Cache rules worth confirming: a BIS hit is never cached (case 1 stays 0 ms because it is a live in-memory lookup), a failure is never cached (repeat case 7 and watch it retry rather than answer instantly), and a free-tier answer never satisfies a request that wants the paid tier.

Backfill

```
npm run enrich:backfill -- --facility HSOP105211 --limit 6
```

Expect all six UNC gastroenterologists as `recovered_in_bis`, with **Neil D Shah — \$26,959 from 5 companies** among them. Add `--write` to store the results. `--missing-email --limit 20` targets the 5,179 physicians with no address; note that sweeping all of them through the paid tier would cost roughly **\$780**, so the defaults are free-tier, 20 rows, dry run.

Promote

```
curl -X POST http://localhost:3002/api/enrich/promote \
  -H 'content-type: application/json' -b <your session cookie> \
  -d '{"npi":"1467521757","email":"nshaheen@med.unc.edu","confidence":98,"source":"web"}'
```

Then confirm the row in `app_contacts` and that the brief's **Contact Intelligence** section shows it, with `source` recording who confirmed it.

4. The edge case worth seeing

```
npm run enrich:try -- agilman@unchealth.org --free    # unresolved (39%)
npm run enrich:try -- agilman@unchealth.org --web     # external, 95% – ANDREW L GILMAN
```

BIS holds **Andrew J Gilman** (NPI 1336534387) at UNC. The web tier, working from a guessed address, confidently resolves **Andrew L Gilman** (NPI 1770673477) — a different middle initial, and `external` rather than `recovered_in_bis`.

Neither answer is obviously wrong from the outside, which is exactly the point: a high confidence score is not certainty. This is why every field carries a source link, why disagreements are shown rather than resolved silently, and why promoting anything into `app_contacts` requires a person to press the button. When two candidates are this close, open the proof URLs before trusting either.

5. Troubleshooting

Symptom	Cause	Fix
Everything returns <code>unresolved</code> , log shows <code>[enrichment:nppes] giving up ... fetch failed</code>	DNS , not the app. <code>npiregistry.cms.hhs.gov</code> intermittently SERVFAIL s on the local router (192.168.31.1) while 8.8.8.8 resolves it fine — seen twice during development	<code>nslookup npiregistry.cms.hhs.gov 8.8.8.8</code> to confirm, then point the router/host at 8.8.8.8 or 1.1.1.1. The agent degrades safely meanwhile and never caches the failure
<code>[enrichment:cache] app_external_profiles not found</code>	Setup SQL not run	Run <code>supabase/enrichment-setup.sql</code> ; caching stays in-process until then
<code>PAID LOOKUP</code> never appears	<code>ANTHROPIC_API_KEY</code> unset, or the caller supplied a name / used a shared mailbox	Check the key; force with <code>--web</code>
Publication count looks implausible	A common surname. The count is narrowed by affiliation and specialty, and labelled " <i>surname match only</i> " when it cannot be	Open the <code>searchTerm</code> in PubMed and check
No enrichment card in the UI	An attendee already matched BIS (nothing to enrich), or the only attendee is the organizer	Check <code>status</code> via <code>npm run enrich:try</code>